

Technology transfer offices and commercialization of university intellectual property: performance and policy implications

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Abstract The recent increase in the rate of commercialization of intellectual property at US and European universities has important performance and policy implications. We review recent studies of the antecedents and consequences of these activities and then draw some lessons learned for policy-makers. A key conclusion is that universities and regions must formulate and implement coherent and feasible technology transfer/commercialization strategies.

Key words: technology transfer offices, entrepreneurship, intellectual property

JEL classification: L31, M13, O31, O32, O33, O34

I. Introduction

In recent decades, almost all research universities in the USA and Europe have established technology transfer offices (henceforth, TTOs) to commercialize their intellectual property (IP). In the USA, the Association of University Technology Managers (AUTM, 2006) reports that the annual number of patents granted to US universities rose from fewer than 300 in 1980 to 3,278 in 2005, while licensing of new technologies has increased almost fourfold since 1991. Annual licensing revenue generated by US universities rose from about \$160 m in 1991 to \$1.4 billion in 2005. In 2005 alone, 628 university-based start-up companies were

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launched, while 5,171 new firms based on university-owned IP have been created since 1980. The pattern observed in the USA is part of an international phenomenon, with marked increases in licensing, patenting, and university-based start-up companies also evident across Europe, Australia, Canada, and elsewhere (Wright *et al.*, 2007). Technologies that have been commercialized via a TTO include the famous Boyer–Cohen ‘gene-splicing’ technique that launched the biotechnology industry, diagnostic tests for breast cancer and osteoporosis, internet search engines (e.g. Lycos), music synthesizers, computer-aided design (CAD), and environmentally friendly technologies.

TTOs serve as an ‘intermediary’ between suppliers of innovations (university scientists) and those who can potentially (help to) commercialize them, i.e. firms, entrepreneurs, and venture capitalists. TTOs facilitate commercial knowledge transfers of IP resulting from university research through licensing to existing firms or start-up companies of inventions or other forms. The activities of TTOs have important economic and policy implications, since licensing agreements and university-based start-ups (spin-offs) can result in additional revenue for the university, employment opportunities for university-based researchers (especially, post-docs) and graduate students, and local economic and technological spillovers through the stimulation of additional R&D investment and job creation.

There is increasing recognition that technology commercialization can occur through several modes. Traditionally, the emphasis has been on licensing and patenting. Thursby and Thursby (2007, this issue) examine the policy issues relating to university licensing. More recently, increased attention has been devoted to the creation of spin-off firms by academic scientists. Scholars have examined university technology commercialization and entrepreneurship, typically focusing on the ‘performance’ of TTOs, while also analysing agents engaged in commercialization, such as academic scientists. Several authors have assessed the determinants and outcomes of faculty involvement in technology commercialization, such as the propensity of academics to patent, disclose inventions, co-author with industry scientists, and form university-based start-ups.

These developments give rise to a number of important policy-related questions.

- (i) What is the role of TTOs in commercializing university IP?
- (ii) How successful are TTOs in generating revenue from the various modes of university IP?
- (iii) What are the key challenges in enabling TTOs to generate revenues from university IP?
- (iv) What policies need to be developed to meet these challenges?

In this paper, we review the burgeoning literature on TTOs and derive some lessons learned for practitioners and policy-makers. The remainder of this article is organized as follows. In section II, we describe the role of TTOs and review some key theoretical studies. Next, we discuss empirical studies on the efficiency of TTOs in commercializing university IP. The evidence on both university licensing and patenting (section III) and university-based start-ups (section IV) is examined. Section V presents lessons learned and policy recommendations, while section VI discusses recent EU policy initiatives in this arena.

II. Theoretical analysis of the role of the TTO

Consistent with Siegel *et al.* (2003a), we conjecture that there are three agents involved in commercialization: university scientists, technology-transfer or licensing officers and/or

other university research administrators, and corporate (venture capital) managers and/or entrepreneurs who (help to) commercialize university-based technologies.

(i) Differences in culture and objectives

In understanding the commercialization process, it is useful to reflect on the incentives and cultures of these three key stakeholders.

Firms and entrepreneurs seek to commercialize university-based IP for profit. When innovation is a key source of competitive advantage, it is critical to maintain proprietary control over these technologies. Therefore, firms typically wish to secure exclusive rights to university-based technologies. Speed is another major area of concern, since firms and entrepreneurs often seek to establish a 'first-mover' advantage.

The *TTO and other university administrators* generally regard themselves as the guardians of the university's IP portfolio, which can potentially generate revenue. Therefore, they are anxious to market university-based technologies to companies and entrepreneurs, although they will often 'hedge', since they do not want to be accused of 'giving away' lucrative taxpayer-funded technologies or because they want to safeguard the 'researchers and the research environment' that generates innovations. This tends to slow down the commercialization process.

Academic scientists, especially those who are untenured, seek the rapid dissemination of their ideas and breakthroughs. This propagation of new knowledge is manifested along several dimensions, including publications in the most selective scholarly journals, presentations at leading conferences, and research grants. The end result of such activity is peer recognition and a concomitant increase in stature within the academic profession, through citations and stronger connections to key social networks in academia.

Faculty members may also seek pecuniary rewards, which can be pocketed or ploughed back into their research to pay for laboratory equipment, graduate students, and post docs. Lacetera (2005) discusses the decision by academic research teams to undertake commercially oriented activities and their subsequent performance, as compared to industrial teams, taking into account the differences in objectives and organizational structure they face. Also Aghion *et al.* (2005) model the specific characteristics of agents belonging to the scientific community, as compared to industrial teams, when discussing the commercialization decision. These models are helpful in explaining differential performance of academic versus non-academic spin-offs and the decision when to license versus spinning off.

(ii) Problems in commercialization of university IP

University scientists are the suppliers of innovations, in the sense that they create new knowledge while conducting (funded) research projects. However, before a university-based innovation can be commercialized, several hurdles must be surmounted. Key issues are whether researchers have sufficient incentives to disclose their inventions, how to induce researchers' cooperation in further development in bringing IP to market and whether it is possible to overcome asymmetric-information problems relating to the value of university inventions.

Disclosure of inventions and ownership of IP

Invention disclosures to the university constitute the critical input in the technology-transfer process. In the USA, the Bayh–Dole Act requires that academics who are being funded on a federal research grant disclose their inventions to the university/TTO. Across Europe, there are notable differences in the ownership of university IP, although it is possible to discern some convergence.¹ Based on extensive interviews with faculty members in the USA, Siegel *et al.* (2004) reported that many faculty members are not disclosing their inventions to their university. Survey research by Thursby *et al.* (2001) confirms this finding. Markman *et al.* (2007) recently documented that many technologies are indeed ‘going out the back door’.

The failure of many academics to disclose inventions to the TTO highlights the problems for officers in eliciting disclosures. Although the Bayh–Dole act stipulates that scientists must file an invention disclosure, this rule is rarely enforced. Instead, the university needs to have proper incentive schemes in place, specifying an adequate share for the inventors in royalties or equity. The importance of this share in securing researchers’ cooperation in technology licensing has been analysed by Macho-Stadler *et al.* (1996), Jensen and Thursby (2001), Lach and Schankerman (2004), and Link and Siegel (2005). All of these models focus on licensing, rather than commercialization through start-ups. Nevertheless, empirical studies of start-up formation by universities have demonstrated the importance of royalty regimes of the university, even on academic spin-off creation rates (e.g. Di Gregorio and Shane, 2003; O’Shea *et al.*, 2005).

Bringing university IP to the market

If a faculty member files an invention disclosure, the TTO (which, in most countries, is the owner of the university IP) decides whether the invention should be patented, usually in consultation with a committee of faculty experts. In making this decision, the TTO typically attempts to assess the commercial potential of the invention. Sometimes, firms or entrepreneurs have already expressed sufficient interest in the new technology to warrant filing a patent. If industry expresses little interest in the technology, universities may be reluctant to file for a patent, given the high cost of filing and protecting patents. When a patent is filed and awarded, the university typically attempts to ‘market’ the invention, by contacting firms that can potentially license the technology or entrepreneurs who are capable of launching a start-up firm based on the technology.

Faculty members may also become directly involved in the licensing agreement as technical consultants or as entrepreneurs in a university spin-off. Jensen and Thursby (2001) show that faculty involvement in the commercialization of a licensed university-based technology increases the likelihood that such an effort will be successful. In order to provide incentives for faculty involvement, licensing agreements should entail either upfront royalties or royalties at

¹ In the UK, universities have increasingly enforced their ownership rights to the IP generated by academic scientists, with the royalties associated with it being distributed between the relevant parties on an institutionally organized basis. Germany and Belgium adopted Bayh–Dole-type legislation in the late 1990s, while in France, this type of regulation had existed for some time. In Italy, public researchers receive the right of ownership of their IP but in most cases the university makes a formal contract on an individual basis to give the IP rights to the university. Discussions are in progress in Sweden and Finland to change to a Bayh–Dole-type arrangement from one where academic scientists owned their IP. Most European countries have changed their legislation to make it possible and/or more attractive to researchers and academics to take equity in a start-up and/or receive royalties. In France, for example, it was illegal before 1999 for an academic to assume equity in a start-up company (for additional discussion of this issue, see Wright *et al.* (2007)).

a later date. For spin-offs, Macho-Stadler *et al.* (2006) demonstrate how the optimal contract between the university TTO, the researcher, and the venture capitalist, entails the allocation to the researcher of founder shares to secure her involvement in the venture. It may also require the researcher to be financially involved in the project as a way to give her incentives to provide effort. The creation and development of a spin-off may generate higher financial returns to universities than licensing. However, we shall see in the next section that TTOs may encounter numerous challenges in pursuing entrepreneurial activities.

Asymmetric information and valuation of inventions

Even when the disclosure and researcher involvement problem is remedied through appropriate incentive schemes, not all potentially viable inventions will be patented and licensed by the university. This relates to the problem of asymmetric information between industry and science on the value of the inventions. Firms typically cannot assess the quality of the invention *ex ante*, while researchers may find it difficult to assess the commercial profitability of their inventions. This problem is studied in Macho-Stadler *et al.* (2007), who use a reputation argument for a TTO to alleviate the asymmetric information problem. The authors demonstrate that larger TTOs may have an incentive to shelve some of the projects in order to build a reputation for delivering good projects, thus raising buyers' beliefs on expected quality. Their results support the importance of a critical size for TTOs to be successful. The authors also predict that establishing a TTO may result in fewer licence agreements, but higher average licence revenues.

(iii) The rationale for TTOs

While in Macho-Stadler *et al.* (2007) the rationale for establishing a TTO is in the pooling of inventions for reputation building, several theoretical papers have provided other rationales for universities to establish TTOs. Jensen *et al.* (2003) model the process of faculty disclosure and university licensing in the USA through a TTO as a game, in which the principal is the university administration and the faculty and TTO are agents who maximize expected utility. The authors treat the TTO as a dual agent, i.e. an agent of both the faculty and the university. Faculty members must decide whether to disclose the invention to the TTO and at what stage, i.e. whether to disclose at the most embryonic stage or wait until it is a lab-scale prototype. The university administration influences the incentives of the TTO and faculty members by establishing university-wide policies for the shares of licensing income and/or sponsored research. If an invention is disclosed, the TTO decides whether to search for a firm to license the technology and then negotiates the terms of the licensing agreement with the licensee. Quality is incorporated in their model as a determinant of the probability of successful commercialization. According to the authors, the TTO engages in a 'balancing act', in the sense that it can influence the rate of invention disclosures, must evaluate the inventions once they are disclosed, and negotiates licensing agreements with firms as the agent of the administration. However, as Clarysse *et al.* (2007) show, the dual agent role of TTOs and their need to convince their own boards of their achievements in attracting investment funds may lead them to over-value projects compared to their underlying quality. All this calls for proper governance and incentive schemes inside the TTO (see also Belenzon and Schankerman, 2007).

Hellman (2005) models the advantage a TTO has, compared to individual scientists or teams, in terms of lower costs of searching for potential buyers, due to specialization and/or

lower opportunity cost of time. He finds that scientists are more likely to delegate their search to TTOs when there is patent protection. Similarly, Hoppe and Ozdenoren (2005) present a theoretical model to explore the conditions under which innovation intermediaries, such as TTOs, emerge to reduce the uncertainty problem. In their model, firms seek to invest in inventions, but they cannot estimate the value of the technology with certainty. Intermediaries are able to make a sunk investment, acquiring the expertise to locate new inventions, sort profitable ones from unprofitable ones, and assess the efficiency level of potential licensees. The authors show that the fixed set-up costs of TTOs can be recovered if the size of the invention pool is large enough to exploit the economies of sharing expertise. The intermediary will reduce the uncertainty problem, although the authors still find a high probability of inefficient outcomes owing to coordination failure.

As noted in Siegel and Phan (2005), TTOs are increasingly encountering a key strategic choice regarding the commercialization of IP: whether to emphasize licensing or spin-offs. Such choices are likely to be determined by the university's perception of the relative financial returns and their desire to generate economic/knowledge spillovers to the community. Significant informational asymmetries and uncertainties regarding the potential markets for innovations may affect this choice. TTOs may have an advantage over academic scientists in identifying opportunities and developing spin-offs through their commercial networks and business development expertise. However, scientists can play a hold-up role if their expertise is necessary for the development of the technology and their preference is for a spin-off rather than a licensing agreement (Lockett *et al.*, 2005). Furthermore, attracting and creating appropriate incentives for TTO personnel, who possess the appropriate set of skills required to develop spin-offs, rather than negotiating licences, can be problematic.

The stark disparities in the motives, perspectives, and cultures of the three key players in this process underscore the potential importance of understanding how organizational factors and institutional policies influence effective university management of IP. In the following two sections of the paper, we review empirical studies of the institutional and managerial practices involved in the commercialization of university IP. The next section considers studies relating to licensing and patents, while section IV addresses start-ups.

III. Effectiveness of TTOs—licensing and patents

In Table 1, we present a review of selected empirical studies of the performance of TTOs with respect to licensing and patenting. Some of these papers attempt to evaluate the productivity of TTOs, based on data measuring the 'outputs' and 'inputs' of university technology transfer (e.g. Siegel *et al.*, 2003a; Thursby and Thursby, 2002). These studies are typically based on a production-function framework, in which a 'best practice' frontier is constructed. The distance from the frontier represents the level of 'technical' inefficiency, or the inability of the organization to generate maximal output from a given set of inputs. Two methods are used to estimate these frontiers, notably data envelopment analysis (DEA) and stochastic frontier estimation (SFE).

Thursby and Kemp (2002) and Thursby and Thursby (2002) employ DEA methods to assess whether the growth in licensing and patenting by universities can be attributed to an increase in the willingness of professors to patent, without a concomitant, fundamental change in the type of research they conduct. The alternative hypothesis is that the growth in technology commercialization at universities reflects a shift away from basic research

Table 1: Selected papers on university TTOs

Author(s)	Data sets	Methodology	Key results
Siegel <i>et al.</i> (2003a)	AUTM, NSF, US census data, interviews	TFP of university licensing —stochastic frontier analysis and field interviews	TTOs exhibit constant returns to scale with respect to the number of licences; increasing returns to scale with respect to licensing revenue; organizational and environmental factors have considerable explanatory power
Link and Siegel (2005)	AUTM, NSF, US census data, interviews	TFP of university licensing —stochastic frontier analysis	Land grant universities are more efficient in technology transfer; higher royalty shares for faculty members are associated with greater licensing income
Friedman and Silberman (2003)	AUTM, NSF, NRC, Milken Institute 'Tech-Pole' Data	Regression analysis—systems equations estimation	Higher royalty shares for faculty members are associated with greater licensing income
Lach and Schankerman (2004)	AUTM, NSF, NRC	Regression analysis	Higher royalty shares for faculty members are associated with greater licensing income
Rogers <i>et al.</i> (2000)	AUTM, NSF, NRC	Correlation analysis of composite technology-transfer score	Positive correlation between faculty quality, age of TTO, and number of TTO staff and higher levels of performance in technology transfer
Thursby <i>et al.</i> (2001)	AUTM, authors' survey	Descriptive analysis of authors' survey/regression analysis	Inventions tend to be disclosed at an early stage of development; elasticities of licences and royalties with respect to invention disclosures are both less than one; faculty members are increasingly likely to disclose inventions
Foltz <i>et al.</i> (2000)	AUTM, NSF	Linear regression	Faculty quality, federal research funding, and number of TTO staff have a positive impact on university patenting
Bercovitz <i>et al.</i> (2001)	AUTM and case studies, interviews	Qualitative and quantitative analysis	Analysis of different organization structures for technology transfer at Duke, Johns Hopkins, and Penn State; differences in structure may be related to technology-transfer performance
Thursby and Kemp (2002)	AUTM	Data envelopment analysis (DEA) and logit regressions on efficiency scores	Faculty quality and number of TTO staff have a positive impact on various technology-transfer outputs; private universities appear to be more efficient than public universities; universities with medical schools less efficient

Table 1: (Continued)

Author(s)	Data sets	Methodology	Key results
Thursby and Thursby (2002)	AUTM, authors' own survey	DEA	Growth in university licensing and patenting can be attributed to an increase in the willingness of professors to patent and license, as well as outsourcing of R&D by firms, not to a shift towards more applied research
Chapple <i>et al.</i> (2005)	UK-NUBS/UNICO Survey-ONS	DEA and stochastic frontier analysis	UK TTOs exhibit decreasing returns to scale and low levels of absolute efficiency; organizational and environmental factors have considerable explanatory power
Carlsson and Fridh (2002)	AUTM	Linear regression	Research expenditure, invention disclosures, and age of TTO have a positive impact on university patenting and licensing

Notes: NSF is National Science Foundation; NRC is National Research Council; TFP is total factor productivity.

towards more applied research. The authors find support for the former hypothesis. More specifically, they conclude that the rise in university technology transfer is the result of a greater willingness on the part of university researchers to patent their inventions, as well as an increase in outsourcing of R&D by firms via licensing.

Siegel *et al.* (2003a) use SFE to pose a different research question: why are some universities more effective at transferring technologies than comparable institutions? Specifically, they attempt to assess and 'explain' the relative productivity of 113 US university TTOs. Contrary to conventional economic models, they found that variation in relative TTO performance cannot be completely explained by environmental and institutional factors. The implication of this finding is that organizational practices are likely to be an important determinant of relative performance.

Based on estimates of their 'marginal product', it appears that technology licensing officers add significant value to the commercialization process. The findings also imply that spending more on lawyers reduces the number of licensing agreements, but increases licensing revenue. Licensing revenue is subject to increasing returns, while licensing agreements are characterized by constant returns to scale.

The authors supplemented their econometric analysis with qualitative evidence, derived from 55 structured, in-person interviews of 100 university technology-transfer stakeholders (i.e. academics, university administrators, and managers/entrepreneurs) at five research universities in Arizona and North Carolina. The field research allowed them to identify IP policies and organizational practices that can potentially enhance technology-transfer performance.

The qualitative analysis identified three key impediments to effective university technology transfer. The first was informational and cultural barriers between universities and firms, especially for small firms. Another impediment was insufficient rewards for faculty

involvement in university technology transfer. This includes both pecuniary and non-pecuniary rewards, such as credit towards tenure and promotion. Some respondents even suggested that involvement in technology transfer might be detrimental to their career. Finally, there appear to be problems with staffing and compensation practices in the TTO. One such problem is a high rate of turnover among licensing officers, which is detrimental to the establishment of long-term relationships with firms and entrepreneurs. Other concerns are insufficient business and marketing experience in the TTO, and the possible need for incentive compensation.

In a subsequent paper, Link and Siegel (2005) find that a particular organizational practice can potentially enhance technology licensing: the *royalty distribution formula*, which stipulates the fraction of revenue from a licensing transaction that is allocated to a faculty member who develops the new technology. The authors find that universities allocating a higher percentage of royalty payments to faculty members tend to be more efficient in technology-transfer activities. Organizational incentives for university technology transfer appear to be important. This finding was independently confirmed in Friedman and Silberman (2003) and Lach and Schankerman (2004), using slightly different methods and data. Jensen *et al.* (2003) provide an extensive survey of the objectives, characteristics, and outcomes of licensing activity at 62 US universities.² They also find that faculty quality is positively associated with the rate of invention disclosure at the earliest stage of idea development and negatively associated with the share of licensing income allocated to inventors. Having appropriate incentives in place appears to elicit more effort from researchers, and also may serve to attract more productive researchers.

Chandler (1962) and Williamson (1975) drew attention to the economics of internal organization. Bercovitz *et al.* (2001) build on this framework to examine what could be a critical implementation issue in university management of technology transfer: the *organizational structure* of the TTO and its relationship to the overall university research administration. They analyse the performance implications of four organizational forms: the functional or unitary form (U-Form), the multidivisional (M-form), the holding company (H-form), and the matrix form (MX-form). The authors note that these structures have different implications for the ability of a university to coordinate activity, facilitate internal and external information flows, and align incentives in a manner that is consistent with its strategic goals with respect to technology transfer.

To test these assertions, they examine TTOs at Duke, Johns Hopkins, and Penn State Universities and find evidence of alternative organizational forms at these three institutions. The authors attempt to link these differences in structure to variation in technology-transfer performance along three dimensions: transaction output, the ability to coordinate licensing and sponsored research activities, and incentive-alignment capability. While further research is needed to make conclusive statements regarding organizational structure and performance, their findings imply that the organizational form of the TTO does matter.

Belenzon and Schankerman (2007) study the use of incentive schemes within the TTO to tie TTO managers' incentives to overall university objectives. They find that bonuses raise licensing income by increasing the quality of transacted inventions, while specifying local development objectives or imposing government constraints on licensing practices of TTOs generally have a negative impact on licensing revenues.

² See Thursby *et al.* (2001) for an extensive description of this survey.

Other papers have focused exclusively on the corporate perspective on formal university technology transfer. Hertzfeld *et al.* (2006) interviewed and then surveyed chief IP attorneys at 54 R&D-intensive US firms concerning IP protection mechanisms related to university patents. They found that firms expressed great difficulty in dealing with university TTOs on IP issues, citing the inexperience of the TTO staff, the TTO's lack of general business knowledge, and its tendency to overstate the commercial value of the patent. The authors reported that firms had decided to by-pass the TTO in some cases and deal directly with the university scientist or engineer.

Most empirical studies of the performance of TTOs have been based on US data. In recent years, several papers based on EU data have been published. Using DEA and SFE methods, Chapple *et al.* (2007) find in a study of 50 UK universities that TTOs exhibit a low level of absolute efficiency in licensing activity and that there appeared to be decreasing returns to scale. These findings indicate that the recent growth in the size of TTOs has not been accompanied by a corresponding growth in the business skills and capabilities of TTO managers. The findings imply a need to upgrade skills and capabilities and to reconfigure TTOs into smaller units, possibly with a regionally based sector focus.

Debackere and Veugelers (2005) explore the case of K.U. Leuven R&D, the technology-transfer organization affiliated with Katholieke Universiteit Leuven (K.U. Leuven) in Belgium, as well as a comparison group of 11 European research universities. They identify numerous factors influencing the management of technology-transfer relationships. Consistent with evidence from the USA (see Link and Siegel, 2005), they find that incentives and organizational practices are important, in terms of explaining variation in relative performance. Specifically, they report that universities allocating a higher percentage of royalty payments to faculty members tend to be more effective in technology transfer. On the organizational side, the authors find that another critical success factor is what they call a 'decentralized management style', which apparently allows the TTO to be much more sensitive to the needs of its stakeholders.

In sum, the extant literature on TTOs suggests that the key impediments to effective university technology transfer tend to be organizational in nature (Siegel *et al.*, 2003b,c). These include problems with differences in organizational cultures between universities and (small) firms, incentive structures including both pecuniary and non-pecuniary rewards, such as credit towards tenure and promotion, and staffing and compensation practices of the TTO itself.

IV. Effectiveness of TTOs: start-ups

Table 2 presents a summary of key studies of TTO effectiveness. TTOs are increasingly focused on the start-up dimension of technology transfer.³ This increase in start-up or spin-off activity has begun to attract considerable attention in the academic literature. Some of these studies use the TTO or university as the unit of analysis, while others focus on individual entrepreneurs. Each aspect is important in understanding the challenges involved in realizing the revenue potential of start-ups.

³ According to the AUTM (2005), the number of start-up firms at US universities rose from 35 in 1980 to 462 in 2004.

Table 2: Selected papers on start-up formation at universities

Author(s)	Unit of analysis	Data/methodology	Key results
Di Gregorio and Shane (2003)	University-based start-ups	AUTM survey/count regressions of the determinants of the number of start-ups	Two key determinants of start-up formation are faculty quality and the ability of the university and inventor(s) to take equity in a start-up, in lieu of licensing royalty fees; a royalty distribution formula that is more favourable to faculty members reduces start-up formation
O'Shea <i>et al.</i> (2005)	University-based start-ups	AUTM survey/count regressions of the determinants of the number of start-ups	A university's previous success in technology transfer is a key determinant of its rate of start-up formation
Franklin <i>et al.</i> (2001)	TTOs and university-based start-ups	Authors' quantitative survey of UK TTOs	Universities that wish to launch successful technology-transfer start-ups should employ a combination of academic and surrogate entrepreneurship
Lockett <i>et al.</i> (2003)	TTOs and university-based start-ups	Quantitative and qualitative surveys of UK TTOs	Universities that generate the most start-ups have clear, well-defined spin-off strategies, strong expertise in entrepreneurship, and vast social networks
Lockett and Wright (2005)	TTOs and university-based start-ups	Survey of UK TTOs/count regressions of the determinants of the number of start-ups	A university's rate of start-up formation is positively associated with its expenditure on IP protection, the business development capabilities of TTOs, and the extent to which its royalty distribution formula favours faculty members
Clarysse <i>et al.</i> (2005)	TTOs and university-based start-ups	Interviews and descriptive data on 50 universities across 7 European countries	Five incubation models identified. Three match resources, activities, and objectives: low selective, supportive, and incubator. Two do not: competence deficient and resource deficient
Markman <i>et al.</i> (2005a)	TTOs and university-based start-ups	AUTM survey, authors' survey/linear regression analysis	The most attractive combinations of technology stage and licensing strategy for new venture creation—early-stage technology and licensing for equity—are least likely to be favoured by the university (owing to risk aversion and a focus on short-run revenue maximization)
Markman <i>et al.</i> (2005b)	TTOs and university-based start-ups	AUTM survey, authors' survey/linear regression analysis	There are three key determinants of time-to-market (speed): TTO resources, competency in identifying licensees, and participation of faculty-inventors in the licensing process
Bercovitz and Feldman (2004)	Medical school researchers at Johns Hopkins and Duke	Determinants of the probability of filing an invention disclosure	Three factors influence the decision to disclose inventions: norms at the institutions where the researchers were trained and the disclosure behaviours of their department chairs and peers

Table 2: (Continued)

Author(s)	Unit of analysis	Data/methodology	Key results
Audretsch (2000)	Entrepreneurs in the life sciences	101 founders of 52 biotech firms/hazard function regression analysis	Academic entrepreneurs tend to be older, more scientifically experienced
Louis <i>et al.</i> (1989)	Faculty members in the life sciences	778 faculty members from 40 universities/regression analysis	Key determinant of faculty-based entrepreneurship: local group norms; university policies and structures have little effect
Lowe and Gonzalez-Brambila (2007)	Faculty members	150 faculty members from 15 universities/regression analysis	Faculty members are more productive researchers than observationally equivalent colleagues <i>before</i> they established their firms. The research productivity of these academics did <i>not</i> decline in the aftermath of their entrepreneurial activity
Zucker <i>et al.</i> (1998b)	Relationships involving 'star' scientists and US biotech firms	Scientific papers, data on biotech firms from the North Carolina Biotechnology Center (1992) and Bioscan (1993)/count regressions	Location of star scientists predicts firm entry in biotechnology
Zucker <i>et al.</i> (2000a)	Relationships involving 'star' scientists and US biotech firms	Scientific papers reporting genetic-sequence discoveries, /count regressions	Collaboration between star scientists and firm scientists enhances research performance of US biotech firms, as measured using three proxies: number of patents granted, number of products in development, and number of products on the market
Zucker and Darby (2001)	Relationships involving 'star' scientists and Japanese biotech firms	Data on biotechnology firms and the Nikkei Biotechnology Directory	Collaboration between star scientists and firm scientists enhances research performance of Japanese biotech firms, as measured using three proxies: number of patents granted, number of products in development, and number of products on the market
Vanaelst <i>et al.</i> (2006)	Start-ups and entrepreneurial team members	Interview data and comparative univariate analysis	Some researchers actively involved in the first phase of the spin-off exit; new members enter, especially those with commercial human capital; some faculty remain with university but work part-time in a technology development role for spin-off

(i) University characteristics and start-up formation

University culture and strategy may influence the extent of start-up activity and resources devoted to its development. Based on a qualitative analysis of five European universities that had outstanding performance in technology transfer, Clarke (1998) concluded that the existence of an entrepreneurial culture at those institutions was a critical factor in their

success. Roberts (1991) finds that social norms and MIT's tacit approval of entrepreneurs were critical determinants of successful academic entrepreneurship at MIT.

Universities that generate the most start-ups have clear, well-defined strategies regarding the formation and management of spin-offs (Lockett *et al.*, 2003). Degroof and Roberts (2004) and Roberts and Malone (1996) find that the optimal university policy is a comprehensive selectivity/support policy for generating start-ups that can exploit ventures with high growth potential. However, such a comprehensive policy is an ideal that may not be feasible, given resource constraints. The authors conclude that while spin-off policies do matter, in the sense that they affect the growth potential of ventures, it may be more desirable to formulate such policies at a higher level of aggregation than the university. Roberts and Malone (1996) conjecture that Stanford generated fewer start-ups than comparable institutions in the early 1990s because the institution refused to sign exclusive licences to inventor-founders.

As with licensing, the organization and use of resources is an important influence on the start-up creation and development process. Based on evidence from 50 European universities, Clarysse *et al.*, (2005) identify three optimal cases where universities can adopt different resource and capability commitments depending upon whether they are seeking to create small numbers of start-ups that become global businesses generating significant capital gains, national level businesses generating revenue streams, or larger numbers of smaller consultancy and service businesses generating local employment. They also identify sub-optimal cases where university TTOs either attempt to undertake commercialization activities beyond the resources they have available or where they fail to develop the competences needed to achieve particular commercialization targets. These findings indicate a need to match universities' objectives for commercialization, and for spin-offs in particular, with appropriate resources and capabilities and with a realistic perspective on the types of successful spin-offs that can be created given a particular university's science and resource base.

The specific expertise and track record of TTOs may also influence the creation and economic development of spin-offs from universities. O'Shea *et al.* (2005) find that a university's previous success in technology transfer is a key explanatory factor of start-up formation. Previous success may depend on the commercial capabilities of TTO staff (O'Shea *et al.*, 2005). Lockett and Wright (2005) assessed the relationship between the resources and capabilities of UK TTOs and the rate of start-up formation at these institutions. They are able to distinguish empirically between a university's resource inputs and its routines and capabilities. Based on count regressions, Lockett and Wright (2005) concluded that there is a positive correlation between start-up formation and the university's expenditure on IP protection, the business development capabilities of TTOs, and the extent to which its royalty distribution formula favours faculty members. These findings imply that universities seeking to create numerous start-ups should devote greater attention to recruitment, training, and development of technology-transfer officers with broad-based commercial skills.

More successful universities also develop vast social networks that help them generate more start-ups. Franklin *et al.* (2001) analyse perceptions at UK universities regarding entrepreneurial start-ups that emerge from university technology transfer. The authors distinguish between academic and surrogate (external) entrepreneurs and 'old' and 'new' universities in the UK. Old universities have well-established research reputations, world-class scientists, and are typically receptive to entrepreneurial start-ups. New universities, on the other hand, tend to be weaker in academic research and less flexible with regard to entrepreneurial ventures. They find that the most significant barriers to the adoption of entrepreneurial-friendly policies are cultural and informational and that the universities generating the most start-ups (i.e. old universities) are those that have the most favourable

policies regarding *surrogate* (external) entrepreneurs. The authors conclude that the best approach for universities that wish to launch successful technology-transfer start-ups is a combination of academic and surrogate entrepreneurship. This would enable universities simultaneously to exploit the technical benefits of inventor involvement *and* the commercial know-how of surrogate entrepreneurs.

Incentives are also important. Lockett *et al.* (2003) found that equity ownership was more widely distributed among the members of the spin-off company in the case of the more successful universities. Di Gregorio and Shane (2003) conclude that the ability of the university and inventor(s) to assume equity in a start-up, in lieu of licensing royalty fees, is a significant determinant of the number of start-ups. They also find that a royalty distribution formula that is more favourable to faculty members reduces start-up formation, a finding that is confirmed by Markman *et al.* (2005a). Di Gregorio and Shane (2003) attribute this result to the higher opportunity cost associated with launching a new firm, relative to licensing the technology to an existing firm.

Faculty quality in terms of generating world-class research of sufficient novelty is also associated with the start-up activity (Di Gregorio and Shane, 2003; O'Shea *et al.*, 2005). We return to faculty quality in the next section.

(ii) Entrepreneurs and academic scientists and start-up formation

Several studies focus on the role of individual scientists and entrepreneurs in the commercialization of IP. Two central issues concern the human capital characteristics of those scientists who become entrepreneurs and the factors influencing the performance of their ventures and the spillovers that arise. Entrepreneurs at universities are different from other entrepreneurs in certain respects. Audretsch (2000) analyses a dataset on university life scientists, in order to estimate the determinants of the probability that they will establish a new biotechnology firm. Based on a hazard function analysis, including controls for the quality of the scientist's research, measures of regional activity in biotechnology, and a dummy for the career trajectory of the scientist, the author finds that university entrepreneurs tend to be older and more scientifically experienced. Other evidence also suggests that the academics developing high-tech ventures are typically key researchers in their fields (Vohora *et al.*, 2004).

Louis *et al.* (1989) analyse the propensity of life-science faculty to engage in various aspects of technology transfer, including commercialization. Their statistical sample consists of life scientists at the 50 research universities that received the most funding from the National Institutes of Health. The authors find that the most important determinant of involvement in technology commercialization was local group norms.

Networks of academic scientists who become entrepreneurs may be important influences on the performance of spin-offs. Mustar (1997) classified spin-offs depending on their cooperation arrangements with other public and/or private bodies and highlighted the relation between the breadth of the social network and the growth trajectory and attrition rate. Nicolaou and Birley (2003) recognized that different embeddedness of academics in a network of ties external or internal to the university may be associated with different growth trajectories.

The seminal papers by Zucker and Darby and various collaborators explore the role of 'star' scientists in the life sciences in the creation and location of new biotechnology firms in the USA and Japan. In Zucker *et al.* (2000), the authors assessed the impact of these university scientists on the research productivity of new US biotech firms. Research productivity is

measured using three proxies: number of patents granted, number of products in development, and number of products on the market. They find that ties between star scientists and firm scientists have a positive effect on these three dimensions of research productivity, as well as other aspects of firm performance and rates of entry in the US biotechnology industry (Zucker *et al.*, 1998a,b).

An important consideration is whether engagement in venture creation comes at the opportunity cost of reduced publications. Lowe and Gonzalez-Brambila (2007) analyse the research productivity of faculty members who formed a start-up company at 15 US universities. These faculty members were more productive researchers than observationally equivalent colleagues *before* they established their firms. More importantly, the authors report that the research productivity of these academics did *not* decline in the aftermath of their entrepreneurial activity.

Based on comprehensive data from the University of Leuven in Belgium, Van Looy *et al.* (2004) find that engagement in entrepreneurial activity is associated with an increase in research output (publications), without affecting the nature of the publications involved. Nevertheless, some faculty who are not star researchers but who are frustrated by lack of academic recognition may also have research contracts with companies that generate possibilities for joint-venture spin-offs.

In Zucker and Darby (2001), the authors examine detailed data on the outcomes of collaborations between 'star' university scientists and biotechnology firms in Japan. Similar patterns emerge, in the sense that they find that such interactions substantially enhance the research productivity of Japanese firms, as measured by the rate of firm patenting, product innovation, and market introductions of new products. However, they also report an absence of geographically localized knowledge spillovers resulting from university technology transfer in Japan, in contrast to the USA, where they found that such effects were strong. The authors attribute this result to the following interesting institutional difference between Japan and the USA in university technology transfer. In the USA, academic scientists typically work with firm scientists at the firm's laboratories. In Japan, firm scientists typically work in the academic scientist's laboratory. Thus, it is not surprising that the local economic development impact of university technology transfer appears to be lower in Japan than in the USA.

The growth of spin-offs may also require different types of skills and investment in human capital. Vanaelst *et al.* (2006) shed light on the important policy issue concerning the evolution of entrepreneurial teams as the spin-off progresses from research to an independent venture. They show that some researchers, who are actively involved in the first phase of the spin-off process, exit and new members enter, especially those with commercial human capital. Exiting members make a career choice to stay with the university, preferring to continue their research, but often having a part-time role providing the spin-off with continuing technological development.

The research on TTOs and start-up formation summarized in sections III and IV underscores the importance of identifying the interests and incentives of those who manage the technology-transfer process. These studies also highlight the importance of human capital (e.g. staffing of TTOs, 'star scientists', and entrepreneurial teams) and the university culture (e.g. the role of department chairs, and entrepreneurs who are employed at these institutions) and group norms.

V. Challenges and policy recommendations

A synthesis of the literature suggests that in order to address the policy questions set out at the beginning of this article concerning whether and how TTOs can be successful in generating additional revenue from commercializing university IP, several challenges must be addressed by university administrators, industry, and policy-makers (e.g. regional or state authorities) in the USA (Siegel and Phan, 2005) and EU (Wright *et al.*, 2007). In this section, we address the stakeholders at university level and their regional policy counterparts. Policy-makers at the EU level are dealt with in section VI.

Universities should adopt a *strategic approach* to the commercialization of IP. First, they must consider a set of key formulation issues involving choices relating to institutional goals and priorities and consequent resource allocation. Given that universities are heterogeneous, in terms of their resource endowments and scientific base, these choices need to reflect such configurations. Establishing priorities also relates to choices regarding technological emphasis for the generation of licensing opportunities, relating to *stage of development*. For instance, proof-of-concept technologies are likely to be more attractive than other technologies if the strategic objective is licensing for cash, since it is relatively easy to compute economic value under this scenario. Furthermore, such technologies can be codified for efficient arm's-length transfer and they are more likely than other technologies to result in a commercial product, without substantial additional research expense.

Resource allocation decisions must also be driven by increasing recognition that universities need to make strategic choices regarding the *mode of commercialization* they wish to emphasize, i.e. licensing, start-ups, sponsored research, or consulting, and other mechanisms of technology transfer that are focused more directly on stimulating economic and regional development, such as incubators and science parks.

University administrators, backed by regional policy-makers may also need to make a strategic choice regarding *technology field of emphasis*. Opportunities for technology commercialization and the propensity of faculty members to engage in technology transfer vary substantially across fields, both between and within the life sciences and physical sciences (Wright *et al.*, 2004). Universities are likely to be differentially endowed in terms of the scope and depth of their research across different disciplines. Where there are not areas of a critical mass of research excellence and TTO expertise within individual universities, there may need to be development of regional *collaborations*.

Universities also need to formulate *IP and patent strategies*. TTOs need to ensure that IP is clean, well defined, and protected before trying to raise commercial interests. This involves costs in terms of recruiting sufficient expertise or paying for external advice. The ownership of IP needs to be resolved. The IP and patent strategy should thus consider what technology is proprietary to the department, which is licensed on an exclusive base, and which parts are licensed on a non-exclusive basis.

The review of the literature has amply demonstrated the potential opportunity costs of commercialization and the need both at a systemic level and within universities to adapt *promotion and remuneration systems* for academics so that commercialization activities are valued.

A major problem also concerns the ability of universities to attract and remunerate TTO personnel with the appropriate human capital *skills* to support the commercialization strategy. The expansion of commercialization activities to include the creation and development of start-ups means that the requisite skills base extends beyond the need to be able to provide

legalistic skills in protecting IP to a requirement to be able to provide opportunity recognition and other commercialization skills.

VI. A role for the EU in policy-making?

The factors identified in the previous section relate mainly to policy competence at regional or national level. The proximity of science and industry facilitates knowledge transfer, which emphasizes the crucial role of regional and local authorities. Many local and regional policy-makers have implemented programmes to stimulate knowledge transfer (see OECD (2001) for an overview). Fully recognizing this subsidiary principle, there is nevertheless scope for policy initiatives at the supra-national level to leverage further these regional/national initiatives.

Policy-makers at EU level have by now firmly recognized the importance of improving knowledge transfer within the EU, motivated by the underperformance of Europe in comparison to the USA in terms of all knowledge-transfer mechanisms from the public research base to industry: patents, licensing, and spin-off creation. The European Commission (EC) has recently launched a policy programme, 'Putting Knowledge into Practice', to help create a European framework for knowledge transfer. This builds further upon ongoing policy initiatives to instigate reforms of universities in the EU. This section briefly reviews and evaluates the main ingredients of EU policy *vis-à-vis* knowledge transfer, focusing on where the supranational EU level can complement and leverage regional and national policy initiatives.

A major scope for policy at EU level is diffusion of best practices. The Community offers various opportunities for regional and transnational policy learning and to foster practical transnational cooperation between regional and national policies. Through the 'Open Method of Coordination' initiative under the coordination of the EU, member states and stakeholders have been exchanging best practices. Other examples are ERAnet and PRO INNO Europe. This has resulted among other things in 'Good Practice Guidelines' which are currently being promoted and further developed and diffused. These Good Practice Guidelines are consistent with the critical areas that have been identified in the previous sections reviewing the academic literature.

In the important area of reforming science institutions and regulations to incentivize and enable researchers to take part in knowledge transfers, including aligning ownership regimes, the EC lacks direct instruments. It nevertheless tries actively to support, coordinate, and diffuse best practices in this area. Also, in certain fields where critical mass needs a large scale for industry–science links, the EU can support the set-up of flagship programmes, which can provide a reference model to experiment with governance mechanisms for industry–science interactions.

Another problem identified in the literature, as noted earlier, is the shortage of experienced knowledge-transfer personnel across Europe. Professional development exists in a limited number of countries, but these programmes are often inadequate. Specific training programmes and initiatives to professionalize knowledge transfer personnel can be provided at the supranational scale. In addition, a number of member states, together with the EC, are looking to create a framework for mutual recognition of qualifications and an EU-wide accreditation scheme. All this will help to create a pan-European market for TTO expertise.

To overcome the inefficient scale of TTOs in Europe, pooling of resources, particularly of certain specialized functions, can help to create a critical mass. Such pooling can address

a single industry sector or a single knowledge-transfer activity. Examples of pooling of knowledge transfer activities exist in some member states. But in some cases, pooling across countries could be envisaged for those scenarios where the scope and volume of activities would justify the establishment of cross-national establishments. The EC is already actively supporting diverse forms of transnational partnering among TTOs and other public and private partners. Support services such as the 'Innovation Relay Centres' network provide assistance for research organizations and industry, with an emphasis on small and medium-sized enterprises (SMEs). They are well placed to encourage transnational knowledge transfers and links between foreign companies and science organizations, helping to create a pan-European market for industry-science transfers.

To bridge the cultural divide that exists between science and industry, mindsets have to change. One effective method to achieve this objective would be to enhance mobility of staff between science and industry. The EC has been an active proponent of such mobility on a pan-European scale, through the Framework Programme 'Marie Curie Industry–Academia Strategic Partnership' scheme. In addition, the EC supports national and regional initiatives in this respect. The new research, development, and innovation (RDI) State Aid Regulations now also allow member states to support knowledge transfer by subsidizing the (temporary) deployment of highly qualified personnel from research organizations to SMEs.

Additional government funding may be needed to address the lack of public and private investment in knowledge transfer. Again, this public funding will mostly be designed at regional/national level. Nevertheless, the EC can also provide some direct support of these regional initiatives. The 'European Structural and Cohesion Funds' programmes identify a range of knowledge and technology-transfer activities that can be covered by EU co-funding. Also the new 'Community Framework for State Aid for Research, Development and Innovation', clarifies and broadens the scope of member-state funding for knowledge-transfer activities to be allowed under state-aid rules.

Finally, it is critical to benchmark commercial activities of research organizations. Such benchmarking, if conducted on the basis of standardized metrics across the EU, would allow universities to compare their own achievements to European and US institutions. Several surveys have been conducted in recent years to provide an international picture of university knowledge-transfer activities, at EU and worldwide level (AUTM, OECD, Association of European Science and Technology Transfer Professionals (ASTP), and Proton). However, none of these surveys so far has the coverage necessary for detailed comparisons across and beyond Europe. More efforts need to be developed at supranational level to identify metrics for assessing knowledge-transfer activities.

Realizing that closer industry–science links across Europe is a cornerstone for the achievement of the Lisbon strategy, the Commission has outlined a policy framework for improving knowledge transfer, which complements national and regional policy initiatives. Left to do in future is monitoring the progress being made on this front, evaluating policy measures, and improving where necessary.

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